

CSA1701-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

3. Water Jug Problem

Aim:

To develop a Python program that solves the Water Jug Problem using Breadth-First Search to find the sequence of pouring steps required to measure a target amount of water.

Algorithm:

1. Define the capacities of Jug 1 and Jug 2, and the target volume.
2. Initialize a queue with the starting state (0, 0) and an empty visited set.
3. Generate all valid operations: fill Jug 1, fill Jug 2, empty Jug 1, empty Jug 2, pour Jug 1 to Jug 2, pour Jug 2 to Jug 1.
4. Use BFS to explore states; avoid revisiting states.
5. If a state containing the target volume is reached, display the solution path.
6. If the queue empties without finding the target, report no solution.

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```
from collections import deque

def water_jug(jug1_cap, jug2_cap, target):
    visited = set()
    queue = deque([(0, 0)])

    while queue:
        path = queue.popleft()
        x, y = path[-1]

        if x == target or y == target:
            return path

        if (x, y) in visited:
            continue
        visited.add((x, y))

        moves = [
            (jug1_cap, y),
            (x, jug2_cap),
            (0, y),
            (x, 0),
            (x - min(x, jug2_cap - y), y + min(x, jug2_cap - y)),
            (x + min(y, jug1_cap - x), y - min(y, jug1_cap - x))
        ]

        for move in moves:
            if move not in visited:
                queue.append(path + [move])

    return None

print("Solving Water Jug (4L, 3L) for Target 2L:")
solution = water_jug(4, 3, 2)

if solution:
    print("Solution Found!")
    for step in solution:
        print(f"Jug1: {step[0]}L, Jug2: {step[1]}L")
else:
    print("No solution exists.")
```

Output:

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```
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr 2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/VISHAL M/Downloads/ai expl.py
Solving Water Jug (4L, 3L) for Target 2L:
Solution Found!
Jug1: 0L, Jug2: 0L
Jug1: 0L, Jug2: 3L
Jug1: 3L, Jug2: 0L
Jug1: 3L, Jug2: 3L
Jug1: 4L, Jug2: 2L
>>>
```

Result:

Thus, the Water Jug Problem solved successfully using Python.